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Delay Analysis of Multi-User URLLC Systems Under Finite Block-length Regime

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***Abstract*—This paper presents a framework designed to evaluate delay violation probabilities in multi-user downlink Ultra-Reliable Low Latency Communication (URLLC) systems. Emphasizing the significance of employing a finite block-length (FBL) transmission model to address transmission errors, our proposed framework assumes infinite user queues at the base station (BS). Delay violation probabilities are assessed with respect to predefined target delays, computed based on infinite queues. To provide insights into essential trade-offs and performance limits, we conduct a comparative analysis of delay violation performances for three widely recognized user scheduling algorithms: round robin (RR), max-cqi, and longest queue (LQ), across diverse network conditions. Simulation results underscore the critical importance of choosing an appropriate scheduling algorithm to effectively meet demanding target delay violation probabilities.**

***Index Terms*—URLLC systems, delay violation probability, finite block-length, multi-user scheduling**

I. Introduction

Establishing a communication network that is both ultra-reliable and low-latency (URLLC) is a difficult task for fifth-generation (5G) wireless networks and beyond. Reliability is typically measured by the probability of successfully transferring a finite amount of data within a certain period, requiring a very low block error rate between 10^{-9} and 10^{-5} [1]. Latency is managed by communication delay, including processing delay, over-the-air transmission delay, and any delay that occurs due to packet re-transmissions when necessary. It can be seen that URLLC is challenging because ultra-high reliability and low latency are required, but these parameters are inversely proportional to each other [2].

Several studies have explored the intricacies and potential of high-performance URLLC systems. The foundation is established by the authors of [3], who provide an accurate approximation of achievable rates based on block-length, error probability, and signal-to-noise ratio (SNR) over an additive white Gaussian noise (AWGN) channel. This reference offers a fundamental understanding of the theoretical underpinnings of URLLC. Several research papers, such as [4], [5], [6], [7] and [8], have proposed strategies to improve URLLC systems. For instance, [4] focuses on reducing delay-violation probability in wireless communication through joint channel-aware and queue-aware scheduling. On the other hand, [5] studied URLLC in scenarios with strict delay constraints, random channel gains, and high SNR. They found a trade-off between latency, service capability, and error probability, characterized by a gain conservation equation. In their recent work, [6] have analyzed the trade-off between service capability, latency, and error probability in multiple-input multiple-output (MIMO) channels with short packets and Finite Block-length (FBL) codes. They came up with a gain conservation equation to describe this trade-off in the high SNR regime. Two research works, [7] and [8], have analyzed the trade-off between error probability and delay violation probability in wireless communication systems. Thus, as demonstrated in this paper or similarly in other relevant studies, incorporating FBL transmission into URLLC system development is crucial for providing a more accurate model that enhances understanding of key trade-offs and considerations.

The study [9] proposes a queue-aware FBL coding scheme for URLLC that uses variable-length coding to improve reliability and optimize latency. They achieve an optimal trade-off between reliability and latency by considering single links, power allocation, and resource constraints. The delay and peak age of information in a communication system investigated at [10]. They propose using FBL coding to determine violation probabilities and suggest optimal block-length values for improved system performances. The paper [11] discusses designing wireless networks for low-latency machine-to-machine (M2M) highlighting the importance of FBL coding for reliable operation with short deadlines for small packets. Similar to these studies, we have focused on the queue-aware transmission scheme under the FBL transmission model, incorporating a framework that takes multi-user schedulers into account.

This paper proposes a framework to evaluate the probability of delay violation for multi-user downlink URLLC systems. The framework considers the finite

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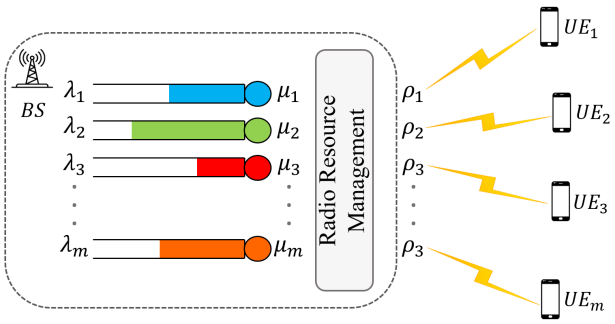


Fig. 1: System model.

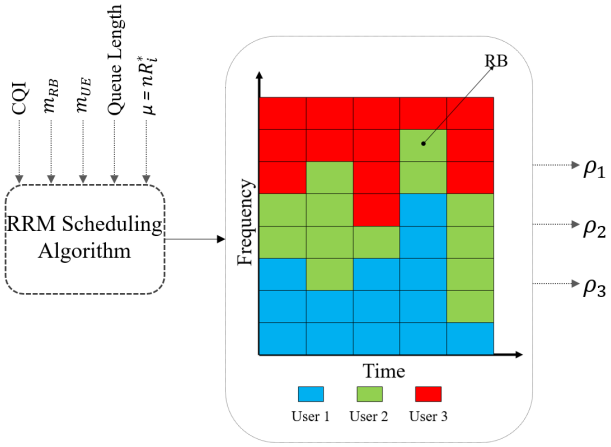


Fig. 2: Radio resource management block.

block-length transmission model, which is critical for considering transmission errors. It evaluates the queuing delay for each bit in the user's queue at the base station (BS) and provides delay violation probabilities based on predefined target delays. The paper also compares the delay violation performances of three well-known user scheduling algorithms under various network conditions. Although the utilized simulation setup has a limited number of users and resource blocks rather than extensive scenarios, results demonstrate the proof of concept for the proposed framework, emphasizing that scheduling algorithm selection is critical to minimize the probability of delay violations.

The paper is structured as follows: Section II introduces the proposed system model, which includes both the physical (PHY) and medium access control (MAC) layers. Section III provides multi-user scheduling algorithms and queuing models. In Section IV, we present the simulation results, and finally, Section V concludes the paper.

II. System Model

In this section, the system model of downlink transmission of a BS to multiple user equipment (UE) is presented. Fig. 1 illustrates that each user has their respective queue at the BS for storing and forwarding incoming data to connected users. This queuing system is commonly analyzed under the assumption of Poisson traffic arrival, a widely adopted model in various research studies for both back-haul and downlink traffic in cellular networks [12]. In line with these previous studies, our proposed model also operates under the assumption of Poisson traffic arrival with the mean of Λ. It generates the random number of λ bits in every time slot. The service rate μ of each queue is determined by the FBL coding. Fig. 2 demonstrates the Radio Resource Management (RRM) block that considers channel information feedback, queue occupancies, number of users m_{UE}, and resource blocks m_{RB}. According to the input parameters, it determines the departure rate ρ with defined scheduling algorithms. It is assumed that users can transmit the SNR acknowledgment information in the same time slot, thus the feedback delay is neglected.

A. Physical Layer

The physical layer of the model defines the data transmission from the BS to the user equipment over a wireless channel. The proposed model is based on discrete time, meaning that the time is divided into time slots with a duration of T second. A flat Rayleigh fading channel model is utilized to model the wireless channel. The fading model assumes that the SNR i^{th} time slot (γ_i) remains the same in each time slot and independently changes from one time slot to another. Thus, the SNR values at the receiver follow independent and identically distributed (i.i.d.) with exponential distribution, $f(\gamma_i) = \frac{1}{\bar{\gamma}} e^{-\frac{\gamma_i}{\bar{\gamma}}}$, where $\bar{\gamma}$ is the average SNR at the receiver, which depends on the utilized path loss model given in Eq. (9). For a time slot i, the spectral efficiency is $C(\gamma_i) = \log_2(1+\gamma_i)$ [13]. Shannon capacity represents the theoretical upper limit of a channel's data transmission capability, while maximal channel coding aims to approach this limit using advanced error-correcting codes. In practice, achieving the Shannon capacity may be challenging due to real-world limitations, such as computational resources and channel impairments. Therefore, maximal channel coding is preferred to optimize data transmission efficiency and reliability within practical constraints.

In [3], the achievable maximal channel coding rate with given block-length n and error probability ϵ is investigated. For this research, the maximal channel coding rate can be closely approximated as:

$$R_i^*(n,\epsilon) \approx C(\gamma_i) - \sqrt{\frac{V(\gamma_i)}{n}} Q^{-1}(\epsilon) \log_2 e \qquad (1)$$

$Q^{-1}(\cdot)$ is the inverse Gaussian Q-function, and $V(\gamma_i)$ denotes the *channel dispersion* [4, Def. 1]:

$$V(\gamma_i) \triangleq 1 - \frac{1}{(1+\gamma_i)^2} \qquad (2)$$

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B. MAC Layer

In the MAC layer, the proposed system model is designed with individual First-in-First-out (FIFO) queues for each user, that are assumed to have an infinite length. These queues store and send the respective packets associated with each user. During each time slot i, first, random $\lambda_{i,m}$ bits, generated by the Poisson traffic, are added to the mth user queue. This paper assumes that we have a Poisson arrival rate, represented with mean Λ_m. Secondly, several data bits are transmitted over the wireless channel with transmission error probability from the queue according to the outputs of the utilized scheduling algorithm at the BS. It is called as a service process $\mu_{i,m}$ in i^{th} time slot for user m. The $\mu_{i,m}$ is $nR^*_{i,m}(n,\epsilon)$ when transmission is successful with a positive acknowledgment, while it is set to zero in case of transmission errors. This implies that bits remain in the queue for subsequent time slots and will be re-transmitted later. The departure process $\rho_{i,m}$ indicates the number of successfully received bits at the destination. Departures are influenced by the queue's backlog and the wireless link's service. Consequently, all data will eventually reach the destination, resulting in a loss-less queueing system. We used the following equations given in [11] to represent the arrival $A(t,\tau)$, service $S(t,\tau)$, and departure $D(t,\tau)$ processes to simulate the delay performances:

$$A(t,\tau) = \sum_{i=\tau}^{t-1} \lambda_{i,m} \tag{3}$$

$$S(t,\tau) = \sum_{i=\tau}^{t-1} \mu_{i,m} \tag{4}$$

$$D(t,\tau) = \sum_{i=\tau}^{t-1} \rho_{i,m} \tag{5}$$

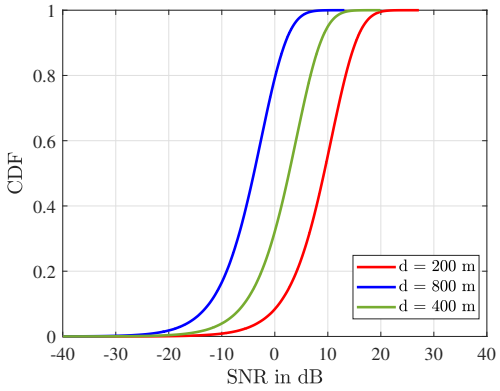


Fig. 3: CDF of SNR for three different positions.

The delay is the count of time slots required for an information bit to arrive at its destination, and it is given in [11] as:

$$W(t) \triangleq inf\{u > 0 : \quad A(0,t) \leq D(0,t+u)\}. \tag{6}$$

A target delay denoted as $\hat{w}$ is introduced. The delay violation probability $p_v(\hat{w})$ represents the probability that the delay exceeds the threshold $\hat{w}$, meaning that there is a likelihood that certain data bits may not be received within a specified deadline. Thus, the delay violation probability for a target threshold of $\hat{w}$ is represented as $p_v(\hat{w}) = \mathbb{P}\{W(t) > \hat{w}\}$.

Introducing a probability of an error in transmission will increase the model accuracy for FBL transmission [11] such that the service model needs to be modified to account for transmission errors. We use the Bernoulli random variable $Z_i \in$ error, success to describe the occurrence of error events:

$$\mu_i = \begin{cases} n\ R^*_i(n,\epsilon), & \text{if } Z_i = success \\ 0, & \text{if } Z_i = error. \end{cases} \tag{7}$$

III. User Scheduling and Queuing Model

A limited number of radio resources are allocated to multiple users at the RRM process in the downlink transmission. The proposed RRM model takes both the physical layer condition of the communication channel and user queue occupancy information to execute a user scheduling algorithm.

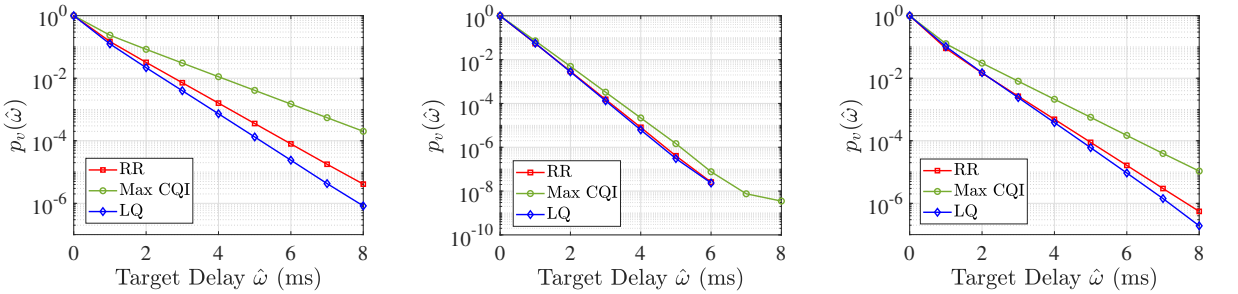
The arrival process for each user is assumed to be an independent Poisson process with λ_m bits/slot. The departure ρ_m and service μ_m processes for individual user queues are determined by the utilized scheduling algorithm in RRM that affects the queue occupancies and the delay violation probabilities. As a variety of user scheduling options can be utilized, the ρ_m and μ_m can be considered to form a general distribution. Each user queue has an independent Poisson arrival process and a general distributed departure process so that the queue model can be interpreted as an M/G/1. The instantaneous queue length of ith time slot is

$$Q[i,m] = (Q[i-1,m] + \lambda_{i,m} - \rho_{i,m})^+, \tag{8}$$

where $(x)^+ = max\{x, 0\}$. Assume that the initial value of the queue $Q[0,m]$ is equal to 0. $Q[i,m]$ denotes the length of the queue at the end of the time slot i for user m. $\rho_{i,m}$ is defined as the departure rate and $\lambda_{i,m}$ indicates the arrival rate.

We have focused on three well-known user scheduling algorithms, namely Round Robin (RR), Max CQI, and Longest Queue (LQ), to utilize in the RRM block. RR Scheduler is an algorithm that is utilized in communication systems to distribute the available bandwidth equitably among several users. This allows multiple users to share the available resources equally aiming to increase

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(a) Scenario 1 d_{UE_1}=200 m, d_{UE_2}=800 m (b) Scenario 2 d_{UE_1}=400 m, d_{UE_2}=400 m (c) Scenario 3 d_{UE_1}=800 m, d_{UE_2}=200 m

Fig. 4: System delay violation probabilities of the scheduling algorithms.

fairness. Max CQI is a scheduling algorithm used in wireless communication systems to allocate bandwidth based on the Channel Quality Indicator (CQI) of each user. In this algorithm, users with higher CQI values are given priority to access the available bandwidth, while users with lower CQI values are allocated at least one resource block. This approach ensures that users with good channel quality receive more bandwidth, which leads to more throughput and higher system capacity [14]. The LQ is a scheduling algorithm used in communication systems to allocate bandwidth based on the length of the queue of each user. In this algorithm, users with longer queues are given priority access to the available bandwidth, while users with shorter queues receive less bandwidth. This approach ensures that users with more data to transmit are served first, leading to a reduction in queue length and an improvement in overall system performance [15].

IV. Simulation Results

In this section, simulation experiments are performed to evaluate three scheduling algorithms, namely RR, MaxCQI, and LQ, under various network scenarios. The downlink transmission system includes two users connected to the BS, and individual user delay violations and system delay violation probabilities are reported as performance results. The reason for defining the total number of users as 2 and the number of resource blocks as 4 in the simulation environment is to save time while obtaining system performance results for different scheduler types and various network scenarios. Since the experiments are performed for 10^8 time slots, increasing these parameters would result in longer simulation times, and hence, we preferred these values to speed up the output generation process. Note that the system delay violation probability is calculated by taking all the transmitted bits into account for all users in the network. Unless otherwise stated, the simulation parameters used in the experiments are given in Table I.

$$PL^{LOS}(dB) = 28 + 22log_{10}(d) + 20log_{10}(f_c) + \chi \quad (9)$$

TABLE I: Simulation Parameters

System parameter	Value
Carrier frequency (f_c)	3.3 GHz
Duration of a time slot (T^{slot})	1 ms
Number of users (m_{UE})	2
Number of Resource Blocks (m_{RB})	4
Λ_1	4320 bits / T^{slot}
Λ_2	6480 bits / T^{slot}
v_1	4.5 km/h
v_2	4.5 km/h
Shadowing fading variance (σ)	2
Noise power (P_N)	-80 dBm
Bandwidth	180 kHz
Transmit power	20 dBm
Radius of Base Station	1 km
Block-length (n)	168
Packet Error Probability (ϵ)	0.0138

User SNR levels are derived using the path loss model described by Eq. 9, operating within a flat Rayleigh fading channel. User mobility adheres to a circular random walk model [16], maintaining a constant speed of pedestrians at 4.5 km/h, which is the mean of the pedestrians, within the BS's coverage area for both users. We categorize users into three channel quality groups near, medium, and far based on their initial distances (d), set to 200 m, 400 m, and 800 m, respectively. Fig. 3 presents the corresponding cumulative distribution function (CDF) of SNR for the three initial user distances. The variations in delay violation probabilities are reported for different scheduling algorithms under different user channel quality scenarios. The system delay violation probabilities of scheduling algorithms are presented in Figure 4 when the initial distances of the user pairs vary. Three different scenarios are created using the user positions mentioned above: near, medium, and far. The first of these scenarios is defined as UE_1 starting to move near to the BS and UE_2 starting to move from a far location point. The system delay violation probability graph of Scenario 1 is given in Figure 4(a). The model we define as Scenario 2 is that both users

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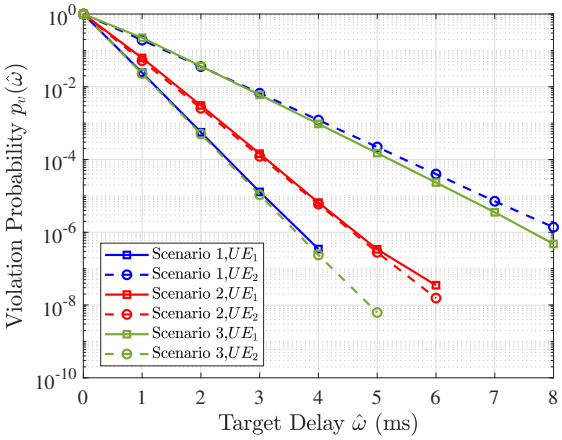


Fig. 5: User delay violation probabilities with LQ scheduling algorithm.

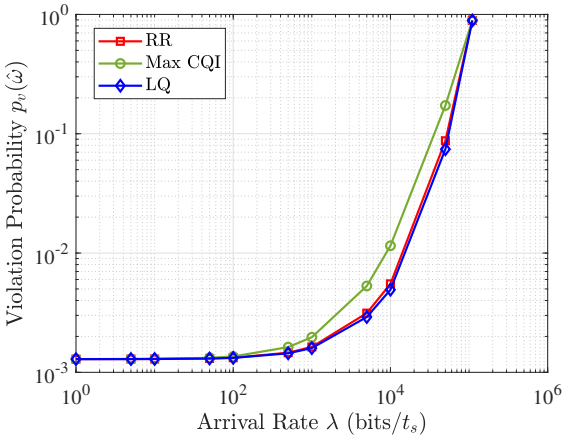


Fig. 6: Scheduling algorithms vs arrival rate.

start their movements from the same point, 400 m, and the system delay violation probability graph for Scenario 2 is shown in Figure 4(b). Finally, Scenario 3 which we defined is shown as UE_1 starting from the far point and UE_2 starting from the near point. The graph for Scenario 3 is given in Figure 4(c). Due to the traffic model, each bit has a unique arrival time. The service of incoming bits may occur at the end of the time slot in which they arrive earliest. Therefore, there is always a delay greater than 0 for each bit. As seen in Fig 4, when the delay violation probability is selected with a target delay of 0, the probability of this situation occurring is 100%. The simulation results show that the LQ scheduler outperforms other schedulers in the given scenarios providing the lowest delay violation probability. One of the fundamental reasons for this is its prioritization of the queue with the increasing load, aiming to reduce the load by increasing the number of resource blocks. As a result, the delay in serving bits is reduced, which in turn has a positive impact on the delay violation probability. It is followed by the RR scheduler and the MaxCQI provides the highest delay violation. As there are no significant differences in the selected user arrival rates, allocating the radio resources equally in the RR scheduler can achieve slightly lower performance results than LQ.

The individual user delay violation probabilities are presented in Fig. 5 when the LQ algorithm is utilized as a scheduler for the three aforementioned scenarios. In scenario 1, since the UE_1 is close to the BS the delay violation probability is better than UE_2. The scenario 3 is defined as the UE_2 in the near region of the BS. Therefore, UE_2 has better delay violation probability. For scenarios 1 and 3, since the arrival rates of users in the low SNR region are different from each other, there are differences between the delay violation probability results. UE_1 gives better results than UE_2 since its arrival rate is lower. The delay violation probabilities are reported in Fig. 6 for three different scheduling algorithms using Scenario 2 as user channel quality conditions when the arrival rate of each user is increased from 1 bit/T^{slot} to 10^6 bits/T^{slot}. When the arrival rate increases, delay violation probabilities increase for all scheduling algorithms until the service capacity is met to serve the arrival traffic. The results show that the queues are not stable and the delay violation probability is close to 1 for the user arrival rates higher than 10^5 bits/T^{slot}. However, there is a specific region where the selection of scheduling algorithms is of critical importance. In studies conducted on the arrival rate values in this region, choosing an appropriate scheduling algorithm would reduce the delay violation probability. In addition, LQ provides the lowest system delay violation probability for all different user traffic arrival rate scenarios when the queues are stable.

V. Conclusion

A comprehensive framework for assessing the probability of delay violation in multi-user downlink URLLC systems is presented in this paper. The proposed framework offers an understanding of queuing delays for each bit in a user's queue at the BS for a finite block-length transmission model. The comparative analysis of three prominent user scheduling algorithms highlights the pivotal role of algorithm selection in minimizing delay violation probabilities across diverse network conditions. The simulation results underscore the significance of tailoring scheduling strategies to specific network scenarios, emphasizing the need for adaptability in the face of varying conditions. In future studies, a system capable of intelligent decision-making, determining the appropriate scheduling algorithm for each UE based on the traffic patterns in the network, could be developed.

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